

Post-Integration Entity-Resolution and Enrichment Assessment

Data Management Project

July 8, 2026

1 Assessment Scope

This report evaluates the automated integration and enrichment steps after cleaning, entity resolution, one-to-one link selection, and unified dataset construction. The assessment uses hand-labeled candidate pairs as a gold standard and measures both classifier error and end-to-end survival into the integrated collection.

Gold Sample Counts

Sample group	Rows
In-calibration rows	600
Out-of-sample rows after CSV de-duplication	300
Out-of-sample rows removed as calibration overlap	12
Out-of-sample rows used	288

2 Metric Definitions

Metric	Formula
MATCH precision	$P_M = TP_M / (TP_M + FP_M)$
Strict MATCH recall	$R_M = TP_M / (TP_M + FN_M)$
Kept MATCH recall	$R_K = K_M / H_M$
Accuracy	$(TP_M + TN_N) / n$
Uncertain rate	U / n
Human MATCH pairs	H_M
Classified MATCH pairs	A_M
Selected MATCH links	L_M
Integrated MATCH pairs	I_M
Link survival	$S_L = L_M / H_M$
Integration survival	$S_I = I_M / H_M$
Dropped by 1:1 selection	D_M
Linked but missing source doc	Q_M
Geocoding error median	$\text{median}(d_i)$
Geocoding error p90	$P_{90}(d_i)$
Geocoding error p95	$P_{95}(d_i)$
Geocoding error max	$\max(d_i)$
Haversine term	$d_i = 2R \arctan 2(\sqrt{a_i}, \sqrt{1 - a_i}), \quad a = \sin^2(\Delta\varphi/2) + \cos \varphi_g \cos \varphi_t \sin^2(\Delta\lambda/2)$
Within 100 m	$\#\{d_i \leq 100\} / m$
Coordinate coverage	$C_T = T_{\text{coord}} / T$
Symbols	H_M : human MATCH pairs; K_M : human MATCH rows predicted MATCH or UNCERTAIN; A_M : human MATCH rows classified MATCH; L_M : linked MATCH pairs; I_M : integrated MATCH pairs; D_M : true matches classified MATCH but not selected; Q_M : selected links missing from integrated; T : Tripadvisor records.

3 Entity Resolution Classifier

The classifier metrics compare the shipped MongoDB candidate labels against the out-of-sample hand-labeled gold rows passed with `-gold-csv`. Manual LLM labels are treated as overrides when present. Labels passed with `-in-calibration-gold-csv` are used for the calibration and cross-validation block in the generated Markdown report, not for the out-of-sample score table.

Metric	Out-of-sample	In-calibration
MATCH precision	96.8%	100.0%
Strict MATCH recall	71.7%	92.0%
Kept MATCH recall	100.0%	100.0%
Accuracy	66.0%	96.7%
Uncertain rate	33.0%	3.3%

4 End-to-End Integration Survival

This section measures whether human-confirmed matches survived beyond classification into selected one-to-one links and integrated restaurant source blocks.

Metric	Out-of-sample	In-calibration
Human MATCH pairs	127	174
Classified MATCH pairs	91	160
Selected MATCH links	88	158
Integrated MATCH pairs	88	158
Link survival	69.3%	90.8%
Integration survival	69.3%	90.8%
Dropped by 1:1 selection	3	2
Linked but missing source doc	0	0

5 Spatial Enrichment

Tripadvisor spatial enrichment is measured by comparing geocoded Tripadvisor coordinates against the Google seed coordinates for human-confirmed matches. Observed distances are truncated by the entity-resolution blocking radius: gross geocoding failures usually appear as missed matches, missing coordinate coverage, or unblockable candidates rather than large distance rows.

Metric	Out-of-sample	In-calibration
Geocoding error median	10.1	10.7
Geocoding error p90	40.5	47.1
Geocoding error p95	52.6	83.4
Geocoding error max	92.7	131.5
Within 100 m	100.0%	97.4%
Coordinate coverage	84.0%	84.0%

6 Reproducibility and Generated Artifacts

The full structured metrics, inspection CSVs, Markdown narrative, and report-ready LaTeX tables are regenerated by `uv run integration-assessment`. This PDF imports the generated tables from the post-integration report folder.